Common Python System Functions (`sys`, `io`, and `os`)

The signatures below use familiar mypy-style types. They are simplified for readability; exact annotations can vary with Python and typeshed versions. `PathLike[str]` means a string-compatible path object.

1 `sys`

`sys.argv` (`list[str]`) contains the command-line arguments. `argv[0]` is usually the script name.

`sys.exit(status: int | str | None = None) -> NoReturn` raises `SystemExit` to end the program; an integer is the exit status.

`sys.path` (`list[str]`) contains the directories and locations Python searches when importing modules.

`sys.stdin` (`TextIO`) is the standard input stream.

`sys.stdout` (`TextIO`) is the standard output stream.

`sys.stderr` (`TextIO`) is the standard error stream.

`sys.platform` (`str`) identifies the current platform, such as "linux" or "win32".

2 `io`

`io.open(file: str | bytes | int | PathLike[str] | PathLike[bytes], mode: str = "r", buffering: int = -1, encoding: str | None = None, errors: str | None = None, newline: str | None = None, closefd: bool = True, opener: Callable[[str, int], int] | None = None) -> IO[Any]` opens a file and returns a text or binary stream, depending on `mode`.

`io.StringIO(initial_value: str = "", newline: str = "\\n") -> StringIO` creates an in-memory text stream.

`io.BytesIO(initial_bytes: bytes = b"") -> BytesIO` creates an in-memory binary stream.

3 `os`

`os.getcwd() -> str` returns the current working directory.

`os.chdir(path: str | PathLike[str]) -> None` changes the current working directory.

`os.listdir(path: str | PathLike[str] = ".") -> list[str]` returns the names of entries in a directory.

`os.mkdir(path: str | PathLike[str], mode: int = 0o777) -> None` creates one directory.

`os.makedirs(name: str | PathLike[str], mode: int = 0o777, exist_ok: bool = False) -> None` creates a directory and any missing parent directories.

`os.remove(path: str | PathLike[str]) -> None` removes a file.

`os.rename(src: str | PathLike[str], dst: str | PathLike[str]) -> None` renames or moves a file or directory.

`os.path.join(path: str | PathLike[str], *paths: str | PathLike[str]) -> str` joins path components using the current platform's path separator.

`os.environ` (`MutableMapping[str, str]`) maps environment variable names to values; changes update the process environment.

`sys.argv`, `sys.path`, the standard streams, `sys.platform`, and `os.environ` are attributes, rather than functions.